Microelectronics Engineering

	Normative 4-year Degree
Major	Microelectronics Engineering 微電子工程學
Degree / Award Title	Bachelor of Engineering in Microelectronics Engineering 工學士 (微電子工程學)
Offering Academic Unit	Department of Electrical Engineering
Normal Period of Study	4 years
Maximum Period of Study	8 years
Credit Units Required for Graduation	121 credit units

Note: The following curriculum information is subject to periodic review and changes.

GE Requirements (31 credit units) Catalogue Term : Semester A 2025/26

		Credit Units
GE University Requirements	GE1401 University English	3
	GE2410 English for Engineering	3
	GE1501 Chinese Civilisation - History and Philosophy	3
	GE1601 Whole-Person Development	1
GE Distributional Requirements	Take at least 1 course from each of the three distributional areas: Area 1: Arts and Humanities Area 2: Study of Societies, Social and Business Organisations Area 3: Science and Technology Visit the GE website for more course information.	12

College-specified Courses ^	MA1200 Calculus and Basic Linear Algebra I *Or* MA1300 Enhanced Calculus and Linear Algebra I	3
	MA1201 Calculus and Basic Linear Algebra II *Or* MA1301 Enhanced Calculus and Linear Algebra II	3
	EE2066 Engineers in Society *Students successfully completed EE4085 Internship: Engineering Practice are not required to take this course.*	3

^ Students are highly encouraged to take MA1200 and MA1201 in which the engineering mathematical contents already fit the needs of the following studies in their majors.

College / School Requirements (6 credit units) **Catalogue Term : Semester A 2025/26**

Course Code	Course Title	Credit Units
CS1302	Introduction to Computer Programming	3
PHY1101	Introductory Classical Mechanics	3

Major Requirements (84 credit units) **Catalogue Term : Semester A 2026/27**

1. Core Courses (66 credit units)

Course Code	Course Title	Credit Units	Remarks
CS2311	Computer Programming	3	
EE1001	Foundations of Digital Techniques	3	
EE1002	Principles of Electrical Engineering	3	
EE1004	Foundations of Information Systems and Data Analysis	3	
EE2000	Logic Circuit Design	3	
EE2004	Microcomputer Systems	3	
EE2005	Electronic Devices and Circuits	3	
EE2104	Introduction to Electromagnetics	3	
EE2800	Semiconductor Physics for Engineers	3	
EE3008	Principles of Communications	3	
EE3070	Design Project	3	
EE3115	Applied Optoelectronic Devices	3	
EE3121	Differential Equations for Electrical Engineering	3	

EE3122	Analogue Circuit Fundamentals	3	
EE3210	Signals and Systems	3	
EE3220	System-on-Chip Design	3	
EE3800	Semiconductor Materials and Devices	3	
EE3801	Microsystems and Nanotechnology	3	
EE4080	Project	6	Students successfully completed **EE4087 Internship: Industrial Project** are not required to take this course. A course to fulfill the internship/consultancy project/research project requirement for normative 4-year degree students.
EE4090	Engineering Training	0	Students successfully completed **EE4085 Internship: Engineering Practice/ FS4007 Start-up Technology Entrepreneur Programme I** are not required to take this course.
GE1354	Introduction to Electronic Design	3	
MA2001	Multi-variable Calculus and Linear Algebra	3	

2. Electives (18 credit units)

Students are required to take at least **SIX** electives from the elective list.

Course Code	Course Title	Credit Units	Remarks
EE3009	Data Communications and Networking	3	
EE3109	Applied Electromagnetics	3	
EE3114	Systems and Control	3	
EE3125	Basic Power Electronics	3	
EE4015	Digital Signal Processing	3	Students can only take either EE4015 or EE5410 to fulfill the curriculum requirement.
EE4035	Optical Fibre Communications	3	
EE4036	Wireless Communications	3	
EE4101	Sustainable Energy Systems	3	
EE4105	Principles of Lasers	3	
EE4107	5G Circuit Design	3	

EE4108	Antennas for Wireless Communications and Sensing Connectivity	3	
EE4142	Introduction to Integrated Photonics	3	
EE4146	Data Engineering and Machine Learning	3	
EE4316	Mobile Data Networks	3	
EE4802	VLSI Circuit Design	3	
EE4803	Analog IC Design	3	
EE4804	Antenna Design for Integrated Circuits	3	
EE5410	Signal Processing	3	P5 level course — Students can only take either EE4015 or EE5410 to fulfill the curriculum requirement.
MSE4171			
SYE4006	Semiconductor Manufacturing and Process Control	3	

Note: Students successfully completed **EE4086** can take one less elective course. See below.

3. Optional Internship Courses

Course Code	Course Title	Credit Units	Remarks
EE4085	Internship: Engineering Practice	3	This internship course of 3CU can be used to fulfill **EE2066 Engineers in Society** and **EE4090 Engineering Training**. A course to fulfill the internship/consultancy project/research project requirement for normative 4-year degree students.
EE4086	Internship: Advanced Topics in Electrical Engineering	3	This internship course of 3CU can be used to fulfill **one elective**. A course to fulfill the internship/consultancy project/research project requirement for normative 4-year degree students.
EE4087	Internship: Industrial Project	6	This internship course of 6CU can be used to fulfill **EE4080 Project**. A course to fulfill the internship/consultancy project/research project requirement for normative 4-year degree students.

Note: Students interested in internship can opt to take EE4085 only, OR EE4086/EE4087 in addition to successful completion of EE4085.

Suggested Study Plan

Curriculum Information Record

Structure and Flowchart